

## Ordinary Differential Equations

## End-Semester Exam

Time: 150 minutes

Total Points: 50

Important

1. Simplify your answers as much as possible and write them in terms of fractions or constants such as  $\sqrt{e}$  or  $\log(4)$  rather than decimals.
2. Show all your work and explain your reasonings clearly.
3. No books, notes, calculators or any papers are allowed. Additional paper is available upon request.
4. You are not allowed to communicate with other students during the exam. Copying will not be tolerated.

1. (10 points) Solve the following:

(a) Find the general solution  $x(t)$  of:  $x' = \frac{2x(x-1)}{t(2-x)}$ .

(b) Solve the initial value problem:  $x'(t) = \cos(tx) - 1$ , with  $x(0) = 0$ .

(c) Find the general solution  $X(t)$  of  $X'(t) = \begin{pmatrix} 2 & 1 \\ 4 & -1 \end{pmatrix} X(t) + \begin{pmatrix} 0 \\ 5t \end{pmatrix}$ .

2. (10 points) Show that there exists a unique function  $x = x(t)$  defined for all  $t > 0$  satisfying the following:

(a)  $x'(t) = \begin{cases} 5-2x & \text{if } x < 0 \\ 3 & \text{if } x \geq 0, \end{cases}$  with  $x(0) = 0$ .

(b)  $x'(t) = \frac{x e^{-t^2}}{1+2t^2+x^4}$ , with  $x(0) = 0$ .

(c)  $x'(t) = \sum_{k=1}^{2024} (1 + \cos^2 x + \sin^4 x)^k$ , with  $x(0) = 0$ .

3. (3 points) Let  $f$  and  $g$  be continuous functions on  $[a, b]$  and suppose  $\phi_1$  and  $\phi_2$  satisfies:

$$x''(t) + f(t)x'(t) + g(t)x(t) = 0 \text{ for } t \in (a, b).$$

Assume  $\phi_1$  attains its maximum and  $\phi_2$  attains its minimum at some  $\xi_0 \in (a, b)$ . Then calculate the Wronskian  $W(\phi_1, \phi_2)(t)$  for all  $t \in (a, b)$ , where

$$W(\phi_1, \phi_2)(t) := \begin{vmatrix} \phi_1(t) & \phi_2(t) \\ \phi_1'(t) & \phi_2'(t) \end{vmatrix}.$$

$$\frac{1}{t^2} = \int \frac{1}{t^2} dt = -\frac{1}{t} = -\frac{ds}{ds}$$

## BASIC ALGEBRA ENDSEM

NOTE:

Marks: 100 weightage 50%

### PROBLEMS

Throughout  $G, H$  are groups. We say  $G$  is a *simple group* if it has no normal subgroups other than  $\{1\}$  and  $G$ . Throughout  $R$  denotes a commutative ring and  $I, J$  are ideals in  $R$ . Furthermore  $M, N$  are  $R$ -modules.

1. Let  $p$  be a prime. Show there exists a *non-abelian* group of order  $p^4$  containing a normal subgroup isomorphic to  $(\mathbb{Z}/p\mathbb{Z})^3$ .  $\cong \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z} \times \mathbb{Z}/p\mathbb{Z}$ . [7]
2. Show a group of order 15 is cyclic. [5]
3. Let  $H$  be a subgroup of a group  $G$ . Construct a bijection from the set of left cosets of  $H$  to the set of right cosets of  $H$ . [5]
4. Prove first Sylow theorem. [7]
5. Let  $R$  be a UFD. Show  $R[X]$  is also a UFD. [11]
6. Prove structure theorem of finitely generated modules over a PID. [11]
7. Let  $\mathbb{Q}_7 = \{x \in \mathbb{Q} \mid x = a/b \text{ where } 7 \text{ does not divide } b\}$ . Show  $\mathbb{Q}_7$  is a local ring. [6]
8. Give an example of ring which is NOT Noetherian. (You have to prove your assertion) [5]
9. Let  $V =$  polynomials in  $X$  of degree  $\leq 3$  with complex coefficients. Let  $T = d/dX$  be the differentiation operator. Find the Jordan form of  $T$ . [6]
10. Let  $R$  be a finite integral domain. Show  $R$  is a field. [6]
11. Let  $p$  be a prime. Let  $m \geq 1$ . Show there exists a field  $F$  with  $|F| = p^n$  and  $n > m$ . [8]
12. Let  $p \equiv 1 \pmod{4}$ . Show  $p$  is NOT a prime in  $\mathbb{Z}[i]$  [6]
13. Let  $R$  be a ring containing  $\mathbb{C}$  as a subring such that  $\dim_{\mathbb{C}} R = n$ . Show  $R$  has at most  $n$  maximal ideals (Hint: Use Chinese remainder theorem) [7]
14. Let  $R$  be a P.I.D. Show if  $p$  is irreducible then it is a prime in  $R$ . [5]
15. Show  $Y^2 + X^2 + 2X$  is irreducible in  $\mathbb{Q}[X, Y]$ . [5]



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(c)  $x'(t) = \sum_{k=1}^{2024} (1 + \cos^2 x + \sin^4 x)^k$ , with  $x(0) = 0$ .

3. (3 points) Let  $f$  and  $g$  be continuous functions on  $[a, b]$  and suppose  $\phi_1$  and  $\phi_2$  satisfies:

$$x''(t) + f(t)x'(t) + g(t)x(t) = 0 \text{ for } t \in (a, b).$$

Assume  $\phi_1$  attains its maximum and  $\phi_2$  attains its minimum at some  $\xi_0 \in (a, b)$ . Then calculate the Wronskian  $W(\phi_1, \phi_2)(t)$  for all  $t \in (a, b)$ , where

$$W(\phi_1, \phi_2)(t) := \begin{vmatrix} \phi_1(t) & \phi_2(t) \\ \phi_1'(t) & \phi_2'(t) \end{vmatrix}.$$

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MA 417, Autumn 2024

4. (3 points) Let  $G: \mathbb{R} \rightarrow \mathbb{R}^2$  be a continuous function such that  $\lim_{t \rightarrow +\infty} \|G(t)\| = 0$  and let  $A$  be a  $2 \times 2$  matrix with real entries such that

$$\max\{\operatorname{Re}(\lambda) : \lambda \text{ is an eigenvalue of } A\} := \sigma < 0.$$

Consider  $\phi(t, X_0)$  satisfying the initial value problem:

$$X'(t) = AX(t) + G(t), \quad X(0) = X_0 \in \mathbb{R}^2.$$

For any  $X_0 \in \mathbb{R}^2$  calculate  $\lim_{t \rightarrow +\infty} \|\phi(t, X_0)\|$ .

Here  $\|Z\| = \sqrt{z_1^2 + z_2^2}$  if  $Z = \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \in \mathbb{R}^2$ .

5. (5 points) Consider the second order ODE depending on the parameter  $\varepsilon > 0$ :

$$x''(t) + \pi^2 x(t) = \cos((\pi + \varepsilon)t).$$

(a) Find the general solution  $x_\varepsilon(t)$  of the above equation depending on the parameter  $\varepsilon > 0$ .

(b) Calculate  $\lim_{\varepsilon \rightarrow 0} |x_\varepsilon(2)|$ .

6. (5 points) Consider the second order ODE:

$$x''(t) = xe^{-x} - 3x \cos^2 x - 2x.$$

(a) Find the equilibrium solutions of the above equation.

(b) Determine the stability behaviour of the equilibrium solutions of the above equation.

7. (5 points) Find all the equilibrium solutions and determine its stability behaviour for the following equations:

(a)  $x'(t) = -3x - x^3$ ,

(b)  $x'(t) = 2x - x^2$ ,

(c)  $x'(t) = -1 + x^2$ .

8. (5 points) Suppose  $\phi(t)$  satisfies:  $\phi'(t) = \cos(1 + \phi^2) \sin(1 - \phi^2)$ . If  $|\phi(0)| \leq 1$ , then show that  $\phi(t)$  is defined for all  $t \in \mathbb{R}$  and  $|\phi(t)| \leq 1$  for all  $t \in \mathbb{R}$ .

9. (4 points) Suppose both  $\phi$  and  $\psi$  satisfy the equation:  $x'(t) = e^{1+\sin x(t)}$  for  $|t| < \pi$ . Show that:

$$|\phi(t) - \psi(t)| \leq |\phi(0) - \psi(0)| e^{te^2} \text{ for all } t \in (-\pi, \pi).$$

te<sup>2</sup>  
e



**Probability I (SI 427)**

Endsem, Full marks: 50

Date: November 22, 2024. Duration: 3 hours

- Use of any unfair means would lead you to FR grade.
- You have to state the theorems clearly if you are using it in your answer.
- You can not refer/recall solved problems done in class/tutorial session to answer any of the questions.

✓ 1. Suppose  $E(X) = E(Y) = 10$ ,  $\text{Var}(X) = 10$ ,  $\text{Var}(Y) = 12$  and  $\text{Cov}(X, Y) = -3$ . Show that  $P(X > Y + 6) < 1/2$ . [3]

✓ 2. Define Borel  $\sigma$ -algebra on  $\mathbb{R}$ . Show that  $\mathbb{Q}$  (set of rational number) is a Borel set. Find  $m([0, 1] \setminus \mathbb{Q})$ , where  $m$  is the Lebesgue measure. [4]

✓ 3. The joint density function of  $X$  and  $Y$  is given by

$$f(x, y) = \begin{cases} k(2 - x - y) & \text{for } x > 0, y > 0, x + y < 2 \\ 0 & \text{elsewhere.} \end{cases}$$

Find  $k$  and  $E[Y|X = 1]$ . ✖ [2 + 2]

✓ 4. Suppose  $X_1 \sim \text{Gamma}(\alpha_1, 1)$ ,  $X_2 \sim \text{Gamma}(\alpha_2, 1)$  and they are independent.

✓ (a) Find joint density of  $(X_1 + X_2, \frac{X_1}{X_1 + X_2})$ . [2]

Recall that  $X \sim \text{Gamma}(\alpha, \lambda)$  if  $f_X(x) = \begin{cases} \frac{\lambda^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\lambda x} & \text{if } x \geq 0 \\ 0 & \text{otherwise.} \end{cases}$

(b) Is it true that  $E\left(\frac{X_1}{X_1 + X_2}\right) = \frac{E(X_1)}{E(X_1 + X_2)}$ ? Justify your answer. [2]

5. ✓ (a) State first Borel Cantelli lemma. [1]

✓ (b) Let  $\{X_n\}$  be a sequence of independent random variables such that  $E(X_n) = 0$  and  $P(|X_n| < M) = 1$  for all  $n$ , where  $M$  is a positive real number. Show that  $\frac{S_n}{n} \xrightarrow{a.s.} 0$ . [4]

6. ✓ (a) State Central limit theorem. [1]

✓ (b) Let  $\{X_n\}$  be a sequence of i.i.d.  $U(-1, 1)$  random variables. Define

$$Z_n = \sqrt{n} \frac{X_1^3 + X_2^3 + \dots + X_n^3}{X_1^2 + X_2^2 + \dots + X_n^2}. \quad N(0, \frac{2}{3})$$

Does  $Z_n$  converge in distribution? Justify your answer. If it converges, find the limit. [4]

7. ✓ (a) Find MGF of Poisson random variable with parameter  $\lambda$ . [1]

- (b) Suppose that the moment generating function of a random variable  $X$  is given by

$$M(t) = \frac{1}{4}(1 + e^{2t} + e^{4t} + e^{6t})e^{e^t + t - 1}, \quad t \in \mathbb{R}.$$

Find  $P(X = 2)$ .

[4]

8. Prove or disprove the following statement:

- (a) Suppose  $X_k \sim N(0, 1)$  for all  $k = 1, 2, 3, 4$  and independent. Then the probability that exactly two of  $X_1, X_2, X_3$  and  $X_4$  are negative is  $3/8$ . [4]

- (b) Let  $\{X_n\}$  be a sequence of random variables such that

$$P(X_n = -3 + 1/n) = 1/3, \quad P(X_n = 0) = 1/3 \text{ and } P(X_n = 3 + 1/n) = 1/3.$$

Then  $X_n$  converges in distribution.

[4]

- (c) Let  $\{X_n\}$  be a sequence of random variables such that

$$P(X_n = -n) = P(X_n = n) = 1/4 \text{ and } P(X_n = 0) = \frac{1}{2}.$$

Then  $X_n$  converges in probability.

[4]

- (d) Let  $X \sim U(0, 1)$ . For  $n = 1, 2, 3, \dots$ , define

$$X_n(\omega) = \begin{cases} 2^{n-1} & \text{if } 0 < X(\omega) < \frac{1}{n}, \\ 0 & \text{otherwise.} \end{cases}$$

Then  $X_n \xrightarrow{a.s.} 0$ .

[4]

- (e) Suppose  $X_n \xrightarrow{a.s.} X$  and  $X_n \xrightarrow{a.s.} Y$ . Then  $P(X = Y) = 1$ .

[4]



Indian Institute of Technology Bombay  
MA 401 Linear Algebra

End-Sem Exam

Date: November 13, 2024

Maximum Marks: 40

Duration: 2 hours

Note:

1. Use results which are covered in the class only.  $\mathbb{R}$  denotes the set of all real numbers and  $M_{n \times n}(\mathbb{R})$  denotes the set of all  $n \times n$  real matrices.
2. Justify all of your answers. Answer without proper justification will be treated as no answer.

1. Let  $A \in M_{4 \times 4}(\mathbb{R})$  be such that the eigenvalues of  $A$  are  $1$  and  $-1$ . Find the determinant of  $A^{1000} - A^{100} + A^{10} - I$ . [3 Marks]

2. Let  $R, L$  be linear maps on  $\mathbb{C}^n$  defined by

$$R(x_1, x_2, \dots, x_n) = (0, x_1, x_2, \dots, x_{n-1}) \text{ and } L(x_1, x_2, \dots, x_n) = (x_2, x_3, \dots, x_n, 0),$$

for all  $(x_1, x_2, \dots, x_n) \in \mathbb{C}^n$ . Find matrix representations of  $R$  and  $L$ . Are  $R$  and  $L$  similar (Justify your answer)? [2 + 1 = 3 Marks]

3. Let  $\mathbb{R}^3$  be the inner product space with usual inner product. Let  $S = \text{Span}\{(1, -1, 0), (1, -1, 1)\}$ . Let  $P$  and  $R$  be the orthogonal projections onto  $S$  and  $S^\perp$  respectively.

- a) Find an orthonormal basis for  $S$  and use it to find  $P(1, 1, 1)$ .
- b) Find an orthonormal basis for  $S^\perp$  and use it to find  $R(1, 1, 1)$ .

4. If  $A$  is a matrix in  $M_{5 \times 5}(\mathbb{C})$  with characteristic polynomial  $(x - 2)^3(x + 5)^2$  and minimal polynomial  $(x - 2)^2(x + 5)$ , what is the Jordan form of  $A$ ? [3 Marks]

5. Find a real  $2 \times 2$  matrix  $A$ , other than  $A = I$  such that  $A^5 = I$ . [4 Marks]

P.T.O.

$$\begin{bmatrix} \begin{pmatrix} 1 & 2 & 0 \\ 1 & 2 \end{pmatrix} & (2) \\ & \begin{pmatrix} -5 \end{pmatrix} \\ & \begin{pmatrix} -5 \end{pmatrix} \end{bmatrix}$$

✓ 6. For each of the following, answer TRUE or FALSE. If the answer is TRUE then provide a proof. Otherwise, provide a counterexample.

✓ a) The set  $\{A \in M_{n \times n}(\mathbb{R}) : 2A - A^T = 0\}$  is a subspace of  $M_{n \times n}(\mathbb{R})$ , where  $A^T$  is the transpose of  $A$ .

✗ b) The set  $\{A \in M_{3 \times 3}(\mathbb{R}) : A(e_1) = (0, 0, 0)\}$  is a subspace of  $M_{3 \times 3}(\mathbb{R})$ , where  $e_1 = (1, 0, 0)$ .

✓ c) If  $A \in M_{2 \times 3}(\mathbb{R})$  and  $B \in M_{3 \times 2}(\mathbb{R})$ , then  $BA$  can not be invertible.

✓ d) If  $A \in M_{2 \times 3}(\mathbb{R})$  and  $B \in M_{3 \times 2}(\mathbb{R})$ , then  $AB$  can not be invertible.

✓ e) If  $A \in M_{2 \times 2}(\mathbb{R})$  and if  $A^2 = A$  then either  $A = 0$  or  $A = I$ .

✓ f) If  $A \in M_{n \times n}(\mathbb{C})$  and if  $A - 2I = -A^2$ , then  $A$  is invertible.

[8 Marks]

7. Let  $A = (a_{ij}) \in M_{n \times n}(\mathbb{C})$  ( $n > 1$ ) be such that rank of  $A$  is 1. Let  $v := (v_1, \dots, v_n) \neq 0$  be a basis for the image of  $A$ .

(a) Show that for all  $i, j = 1, \dots, n$ ,  $a_{ij} = v_i w_j$  for some vector  $w = (w_1, \dots, w_n) \neq 0$ .

(b) How the eigenspace of  $A$  corresponding to the eigenvalue 0 is related to  $w$ ?

③

[5+3 = 8 Marks]

✓ 8. Let  $A \in M_{3 \times 3}(\mathbb{C})$ , and let  $S \in M_{3 \times 3}(\mathbb{C})$  be invertible. If  $SA = -AS$ , show that 0 is an eigenvalue of  $A$ .

[3 Marks]

✓ 9. Let  $T \in M_{n \times n}(\mathbb{C})$  be a self-adjoint matrix. If trace of  $A^2$  is zero, show that  $A = 0$ .

[3 Marks]

$$A^* = A$$

$$AA^* = A^2$$

$$\text{tr}(AA^*) = \text{tr}(A^2) = 0$$

$$\sum_{i=1}^n \sum_{j=1}^n a_{ij} \overline{a_{ij}} = 0$$

$$a_{ij} \overline{a_{ij}} = |a_{ij}|^2$$

$$a_{ij} \overline{a_{ij}} = |a_{ij}|^2$$



Autumn 2024

16-11-2024

MA 403 Real Analysis, End-semester Exam.- part 1  
Duration: 1 hrs 25 min, Total marks: 22

1. (a) State completeness axiom (1)
- (b) Show that a convergent sequence in a metric space has a unique limit point. (1.5)
- (c) Let  $\{x_n\}_{n=1}^{\infty}$  be a sequence such that  $\mathbb{Q} \cap [0, 1] = \{x_n : n \in \mathbb{N}\}$ . Can  $\{x_n\}_{n=1}^{\infty}$  be an increasing sequence? Justify. (1.5)
- (d) Compute the  $\liminf_{n \rightarrow \infty} a_n$  and  $\limsup_{n \rightarrow \infty} a_n$  where  $a_n = 3 + (-1)^n \left( \frac{n+5}{n+6} \right)$  (1)

2. (a) Determine all values of  $\alpha \in \mathbb{R}$  for which the following series converges:  $\sum_{n=1}^{\infty} \left( 1 - n \sin \frac{1}{n} \right)^{\alpha}$  (2.5)

- (b) Let  $a \in \mathbb{R}$ . Is the series  $\sum_{n=1}^{\infty} (-1)^n \sin \frac{a}{n}$  convergent? Is the series absolutely convergent? Justify. (3)

3. Prove that  $\sum_{n=1}^{\infty} \frac{x}{n^{\alpha}(1+nx^2)}$  converges uniformly on  $\mathbb{R}$  if  $\alpha > \frac{1}{2}$ . (3.5)

4. Determine the set of all the points  $x \in \mathbb{R}$  for which the power series converges:  $\sum_{n=1}^{\infty} \left( \frac{2 + (-1)^n}{5 + (-1)^{n+1}} \right)^n x^n$  (3)

5. If  $f : [a, b] \rightarrow \mathbb{R}$  is an integrable function, then prove that function  $|f|$  is integrable. (5)

\*\*\*\*\* End \*\*\*\*\*

$L(f) = U(f)$

Weierstrass M-test

$U(p, 1+x)$

$\sin \frac{a}{n}$

$\frac{n}{n^2(1+nx^2)}$

$\frac{1}{2n^{2+\frac{1}{2}}}$

$\frac{1}{2+\frac{1}{2}} > 1$

$\frac{1}{2} > \frac{1}{2}$

Autumn 2024

16-11-2024

MA 403 Real Analysis,

End-semester Exam.- part 2

Duration: 1 hrs 25 min, Total marks: 23

1. Assume  $f : S \rightarrow T$  is uniformly continuous on  $S$ , where  $S$  and  $T$  are metric spaces. If  $\{x_n\}_{n=1}^{\infty}$  is any Cauchy sequence in  $S$ , prove that  $\{f(x_n)\}_{n=1}^{\infty}$  is a Cauchy sequence in  $T$ . (3)

2. Determine all values of  $s \in \mathbb{R}$  for which the following integral converges:

$$\int_0^1 x^{s-1} e^{-x} dx$$

(4)

3. (a) For a function  $f : [0, 1] \rightarrow \mathbb{R}$  let  $B_n(f)$  denote the Bernstein polynomial of  $f$ . For  $f_2(x) = x^2$ , show that

$$B_n(f_2)(x) = \left(1 - \frac{1}{n}\right)x^2 + \frac{1}{n}x$$

(4.5)

- (b) State Fejer theorem.

(1.5)

4. (a) Define function of bounded variation. (1.5)

- (b) Is the function  $f : [0, 1] \rightarrow \mathbb{R}$ , defined as  $f(x) = x^2 \sin(\frac{1}{x})$  if  $x \neq 0$  and  $f(0) = 0$ , of bounded variation? Justify. (3.5)

5. Assume that all zeros of a polynomial  $P$  of degree  $n \geq 3$  are real and write

$$P(x) = (x - a_1)(x - a_2) \dots (x - a_n),$$

where  $a_i \leq a_{i+1}$ ,  $i = 1, \dots, n-1$ . Show that there exist  $c_i$  such that

$$P'(x) = n(x - c_1)(x - c_2) \dots (x - c_{n-1}),$$

where  $a_i \leq c_i \leq a_{i+1}$ ,  $i = 1, \dots, n-1$ . Show that if

$$Q(x) = (x - a_1)(x - a_2) \dots (x - a_{n-1}),$$

$$Q'(x) = (n-1)(x - d_1)(x - d_2) \dots (x - d_{n-2}),$$

where  $a_i \leq d_i \leq a_{i+1}$ ,  $i = 1, \dots, n-2$ , then  $d_i \geq c_i$ ,  $i = 1, \dots, n-2$ . (5)

\*\*\*\*\* End \*\*\*\*\*